793B Workshop Test

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

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1.0 Introduction

Our client asked us to overhaul 9 793B trucks for its fleet. The process included disassembling trucks in the Cajamarca, Peru mine site and assembling the repaired in the Arequipa, Peru mine site. Additionally, the client requested conditions such as: Cab and chassis change, enclosed shop, scheduled delivery dates, and conditioning of some accessories as per the client's standard, among others. For this purpose it was also decided to repair the components, perform the preassembly and testing in the Ferreyros main shop at Lima. Finally, the assembly was performed in Arequipa.

The main concern is to meet the repair delivery date and quality. According to our experience, problems are normally found during the commissioning due to defects in repaired trucks from other dealers.

Quality control is needed to ensure the quality of the repair, thus we decided to perform the evaluation of most systems by testing without load in the workshop before delivery to mine site in Arequipa.

For the first truck, the differential assembly was installed only partially because the crane is limited to 10 ton max. Since the supply of oil needed to lubricate the differential is approximately 235 gallons, it was decided to remove the final drives, disconnect the shaft transmission, bypass the cooling lines, brake, and parking line in order to test without oil in the differential. Due to this, additional equipment and (man hours) were contracted to finish on time.

For this reason it was proposed fabricate a transmission stand, and to test without the differential assembly. Finally we designed and fabricated an adapter for the transmission test purposes.

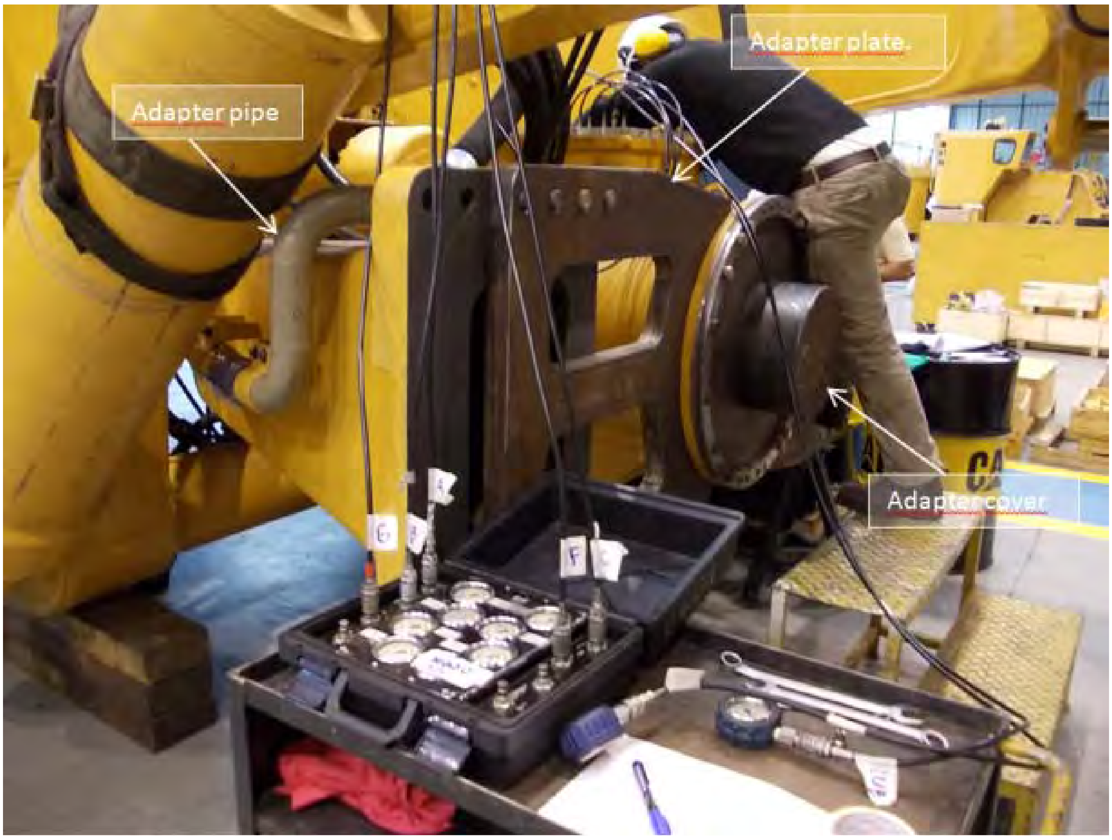
This procedure is accomplished by testing the truck in the workshop with most of components connected. This procedure also eliminated the differential assembly rework, redos, common repair failover, and assured proper operating of the equipment. Therefore equipment delivery was improved with the best delivery date.

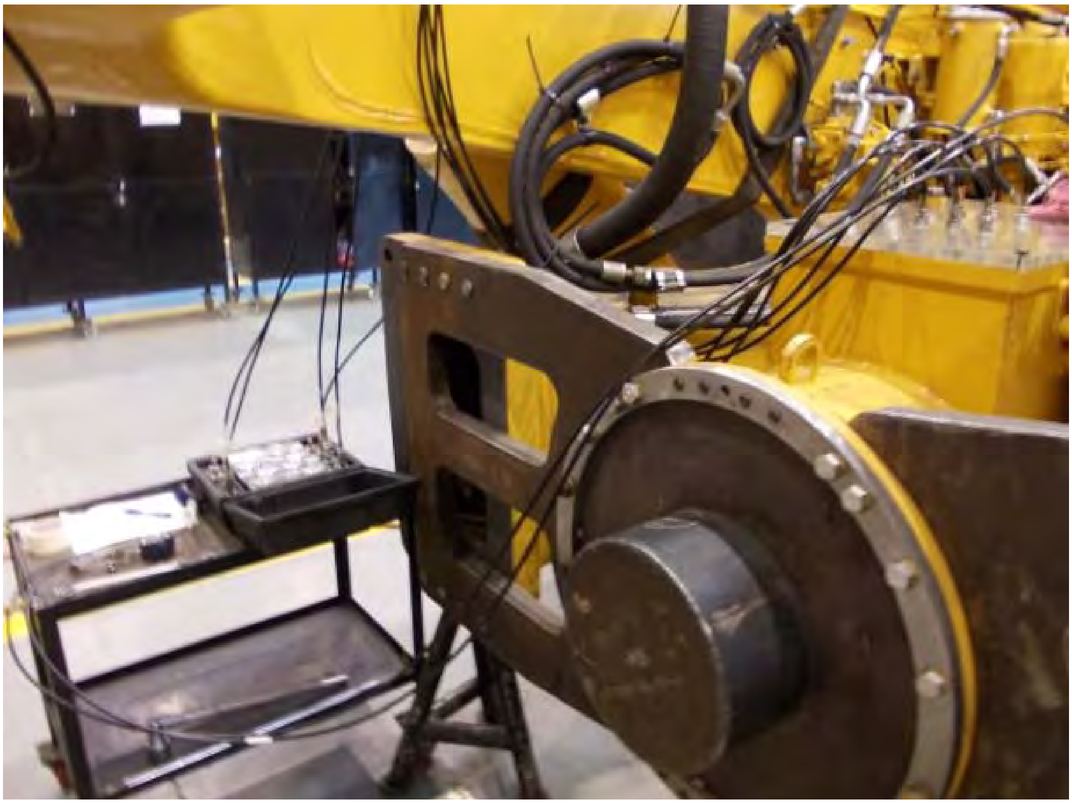
2.0 Best Practice Description

In order to test the truck, a locally fabricated support group is used. It allows support of the transmission assembly while connected to the engine, which is mounted to the truck frame. Brake cooling lines are bypassed and brake oil lines are blocked.

The support group was designed and manufactured in the dealership facilities. It consists of two adapter plates, an adapter cover assembly, and a pipe bypass.

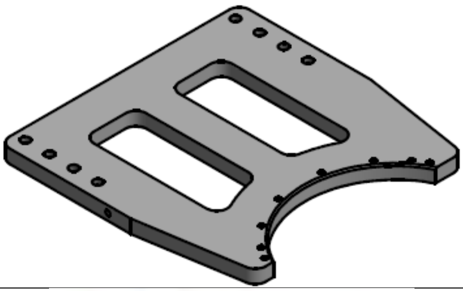
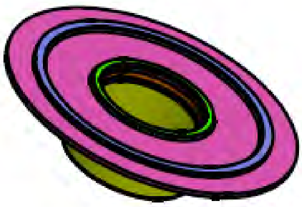
Image of the truck test with the transmission mounted in the device.

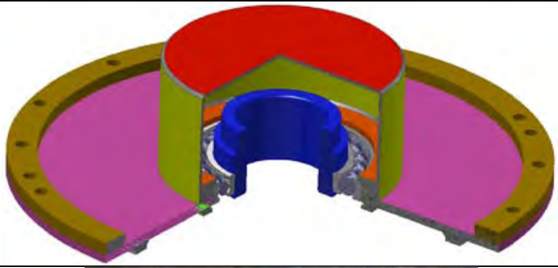




3.0 Implementation Steps

Blue print details for housing reconstruction were used, and support group dimensions were designed by shop supervisor.

Adapter plate	
Adapter cover	
Coupling	

Adapter cover assembly	
Adapter pipe	
Installation verification	

4.0 Benefits

- Reduce total delivery time.
- Eliminate crane & labor costs to remove/install the differential for testing purposes
- The pre-assembled truck is tested and calibrated in the shop, and corrections are made before delivery to mine site
- Dynamic test can be made
- Test mode (without final drives) can be made
- Avoid using 235 gal of SAE 50W oil only for power train testing & adjuster

Image of ET after test.

Parameter	Value
Id equipo	1HL00246
Description of personality module	793B (1HL1-up)

Code	Description
3516B 793 (KNS00677)	
No active codes	
VIMS Principal 793	
839- 4	Presure sensor front right: Voltage under normal
Produck link	
No active codes	
Transmission 793	
No active codes	
Brake 793	
No active codes	

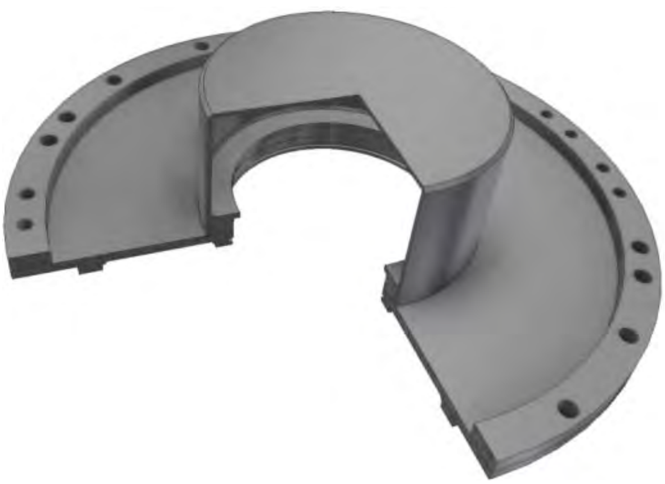
5.0 Resources Required

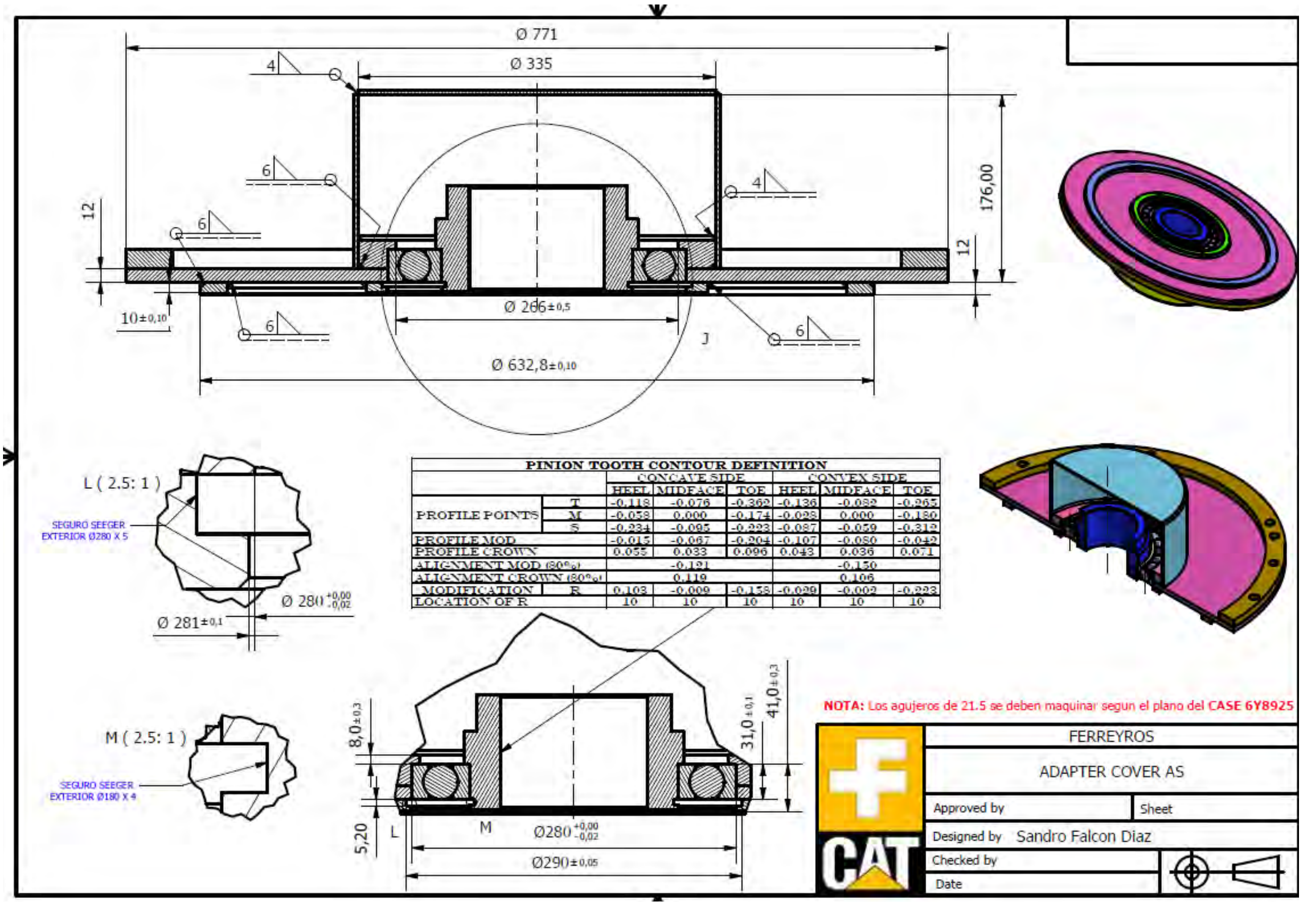
- Weld/fabrication shop
- Estimated investment (material, labor and parts) US $5,500.00.

6.0 Supporting Attachments / References

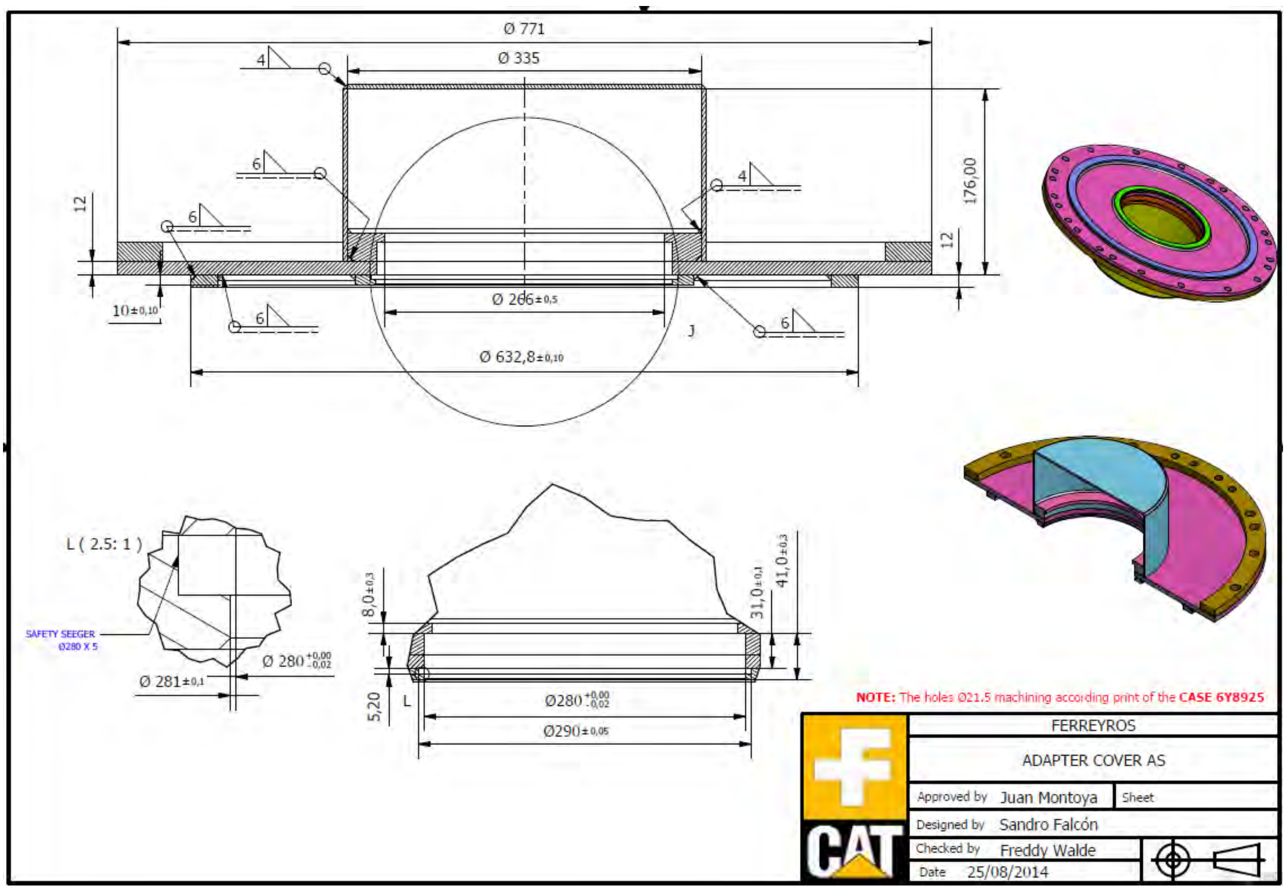
- Attached manufacturing drawings.
- 16036 rigid ball bearing
- 280 mm Øin Seeger lock
- 180 mm Øext Seeger lock
- (01) Seal 5D5428 (for adapter cover assembly)
- (24) bolt 7X2564 (for adapter cover assembly)
- (24) Washer hard 8T3282 (for adapter cover assembly)
- (16) Bolt 9X6152 (for adapter plate)
- (16) Washer hard 3S6162 (For adapter plate)

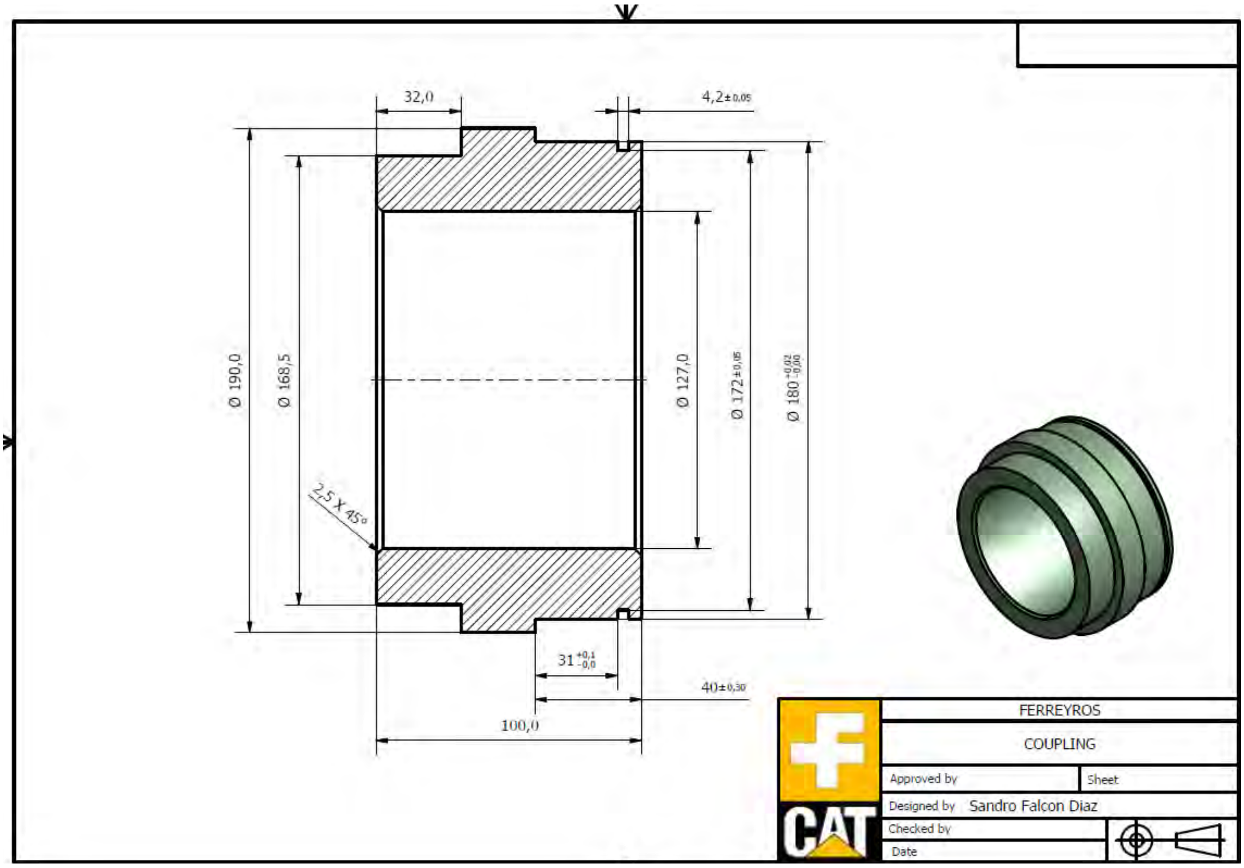
ADAPTER COVER.

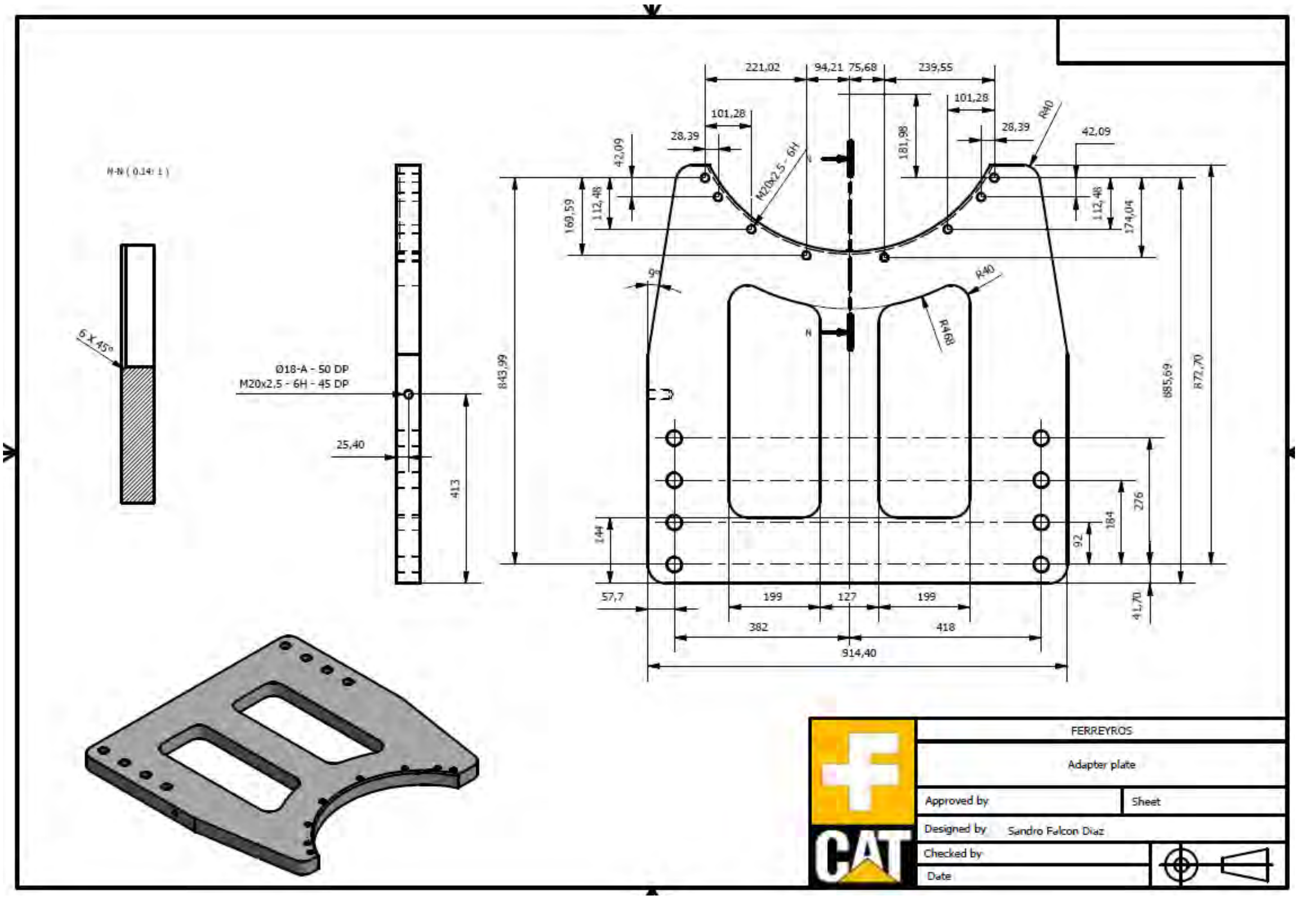


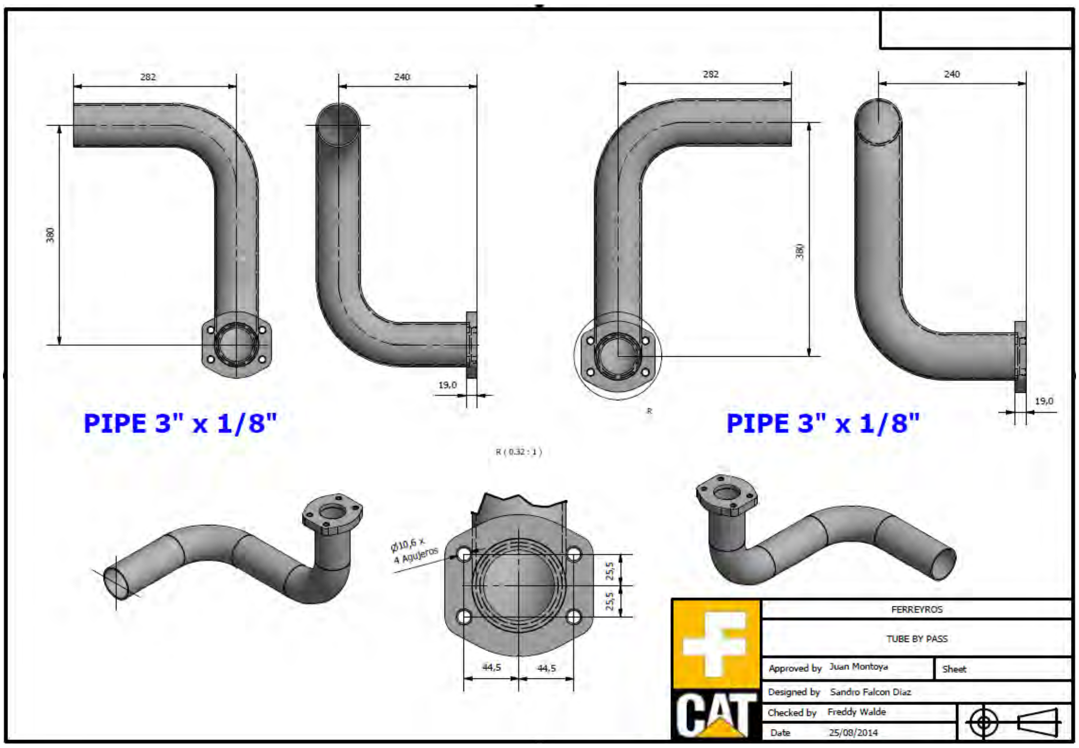


PINION TOOTH CONTOUR DEFINITION							
		CONCAVE SIDE			CONVEX SIDE		
		HEEL	MIDFACE	TOE	HEEL	MIDFACE	TOE
PROFILE POINTS	T	-0.118	-0.076	-0.362	-0.136	-0.082	-0.265
	M	-0.058	0.000	-0.174	-0.028	0.000	-0.180
	S	-0.234	-0.095	-0.223	-0.087	-0.059	-0.312
PROFILE MOD		-0.015	-0.067	-0.204	-0.107	-0.080	-0.042
PROFILE CROWN		0.055	0.033	0.096	0.043	0.036	0.071
ALIGNMENT MOD (80%)			-0.121			-0.150	
ALIGNMENT CROWN (80%)			0.119			0.106	
MODIFICATION	R	0.103	-0.009	-0.158	-0.029	-0.002	-0.223
LOCATION OF R		10	10	10	10	10	10




32,0
4,2±0,05
Ø 190,0
Ø 168,5
Ø 127,0
Ø 172±0,05
Ø 180
2,5 X 45°
31
40±0,30
100,0
FERREYROS
COUPLING
Approved by
Sheet
Designed by Sandro Falcon Diaz
Checked by
Date


221,02
94,21 75,68
239,55
101,28
28,39
42,09
181,98
6 X 45°
Ø18-A - 50 DP
M20x2.5 - 6H - 45 DP
25,40
413
843,99
169,59
112,48
174,04
885,69
872,70
R40
R468
144
57,7
199
127
382
418
914,40
92
184
276
41,70
FERREYROS
Adapter plate
Approved by
Sheet
Designed by Sandro Falcon Diaz
Checked by
Date



7.0 Related Best Practices

None

8.0 Acknowledgements

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Blue Print drawing

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